Exp 03

Cisco packet Tracer

Aim:

To study the cisco packet tracer simulation, tool, understand its user interface design a simple network and analyze the behaviour of hub, switch and peer to peer.

Cisco packet tracer

It is a network simulation and modeling tool. It allows us to create and test complete networks without using real networking equipment we can add PCs, switches, hubs, routers and other devices and observe how packets travel between them.

1) Peer - peer:

Two computers communicate directly A peer to peer network was created by connecting 2 PCs directly IP addresses were assigned to both PCs in the same network. The connectivity btw 2 PC was tested using the ping command executed.

2) Network using HUB.

Connected multiple computers forwards recived date to all parts.

A network was created by connecting 3 PCs to a hub using copper straight to all PCs. cables. when data was sent from one PC to another hub forwarded the packet to all connected hub forwarded the packet to all ports.

3) Networking Switch

Connects Computer & forward data towards the required destination.

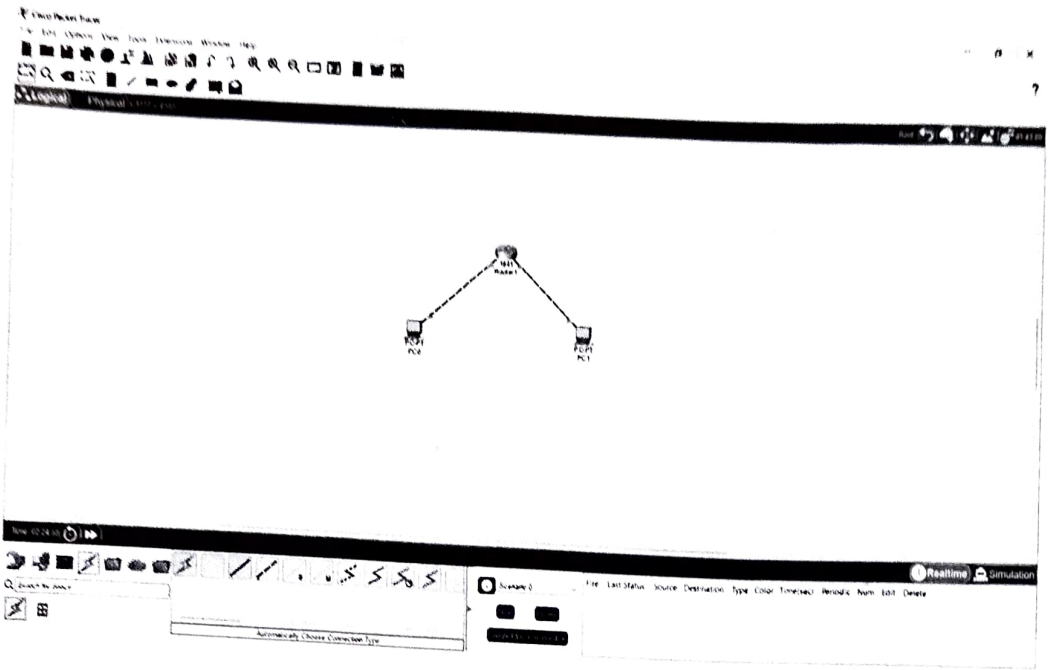
A network was created by connecting 3PCs to a Switch using copper straight through cables. IP addresses were assigned to all PCs and Connectivity was tested using ping Command.

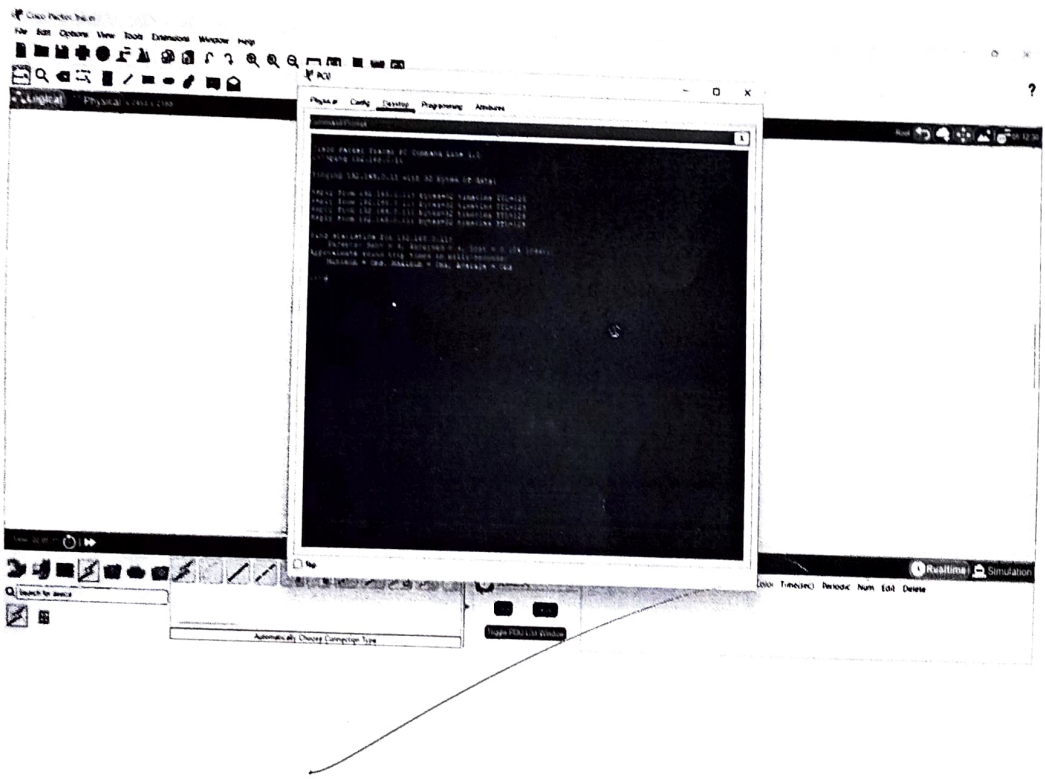
A) Hub & Switch:

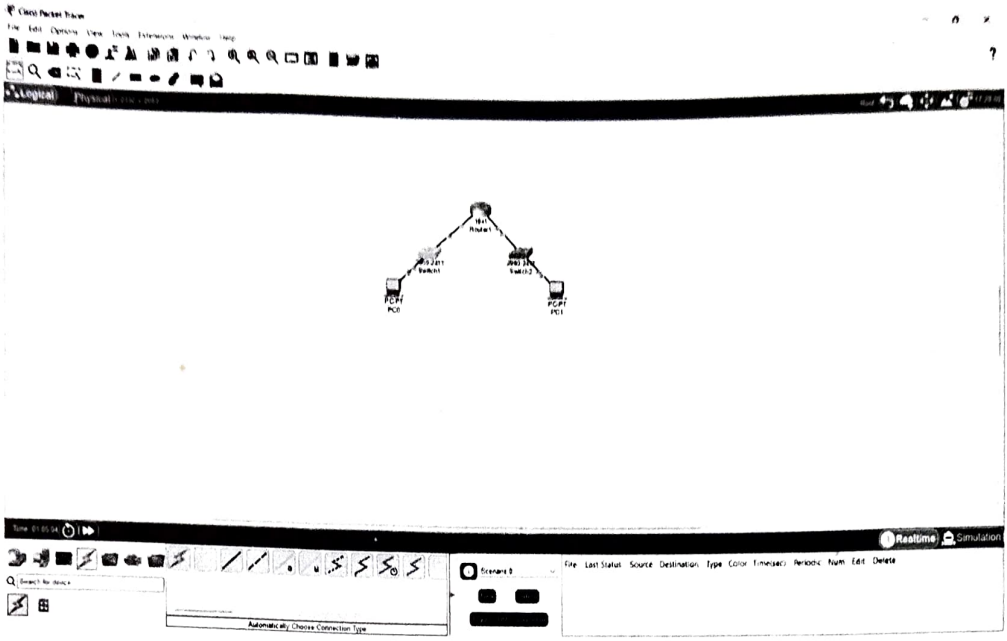
Both hub and switch user used in the network the packet forwarding behaviour of both devices was observed & assigned.

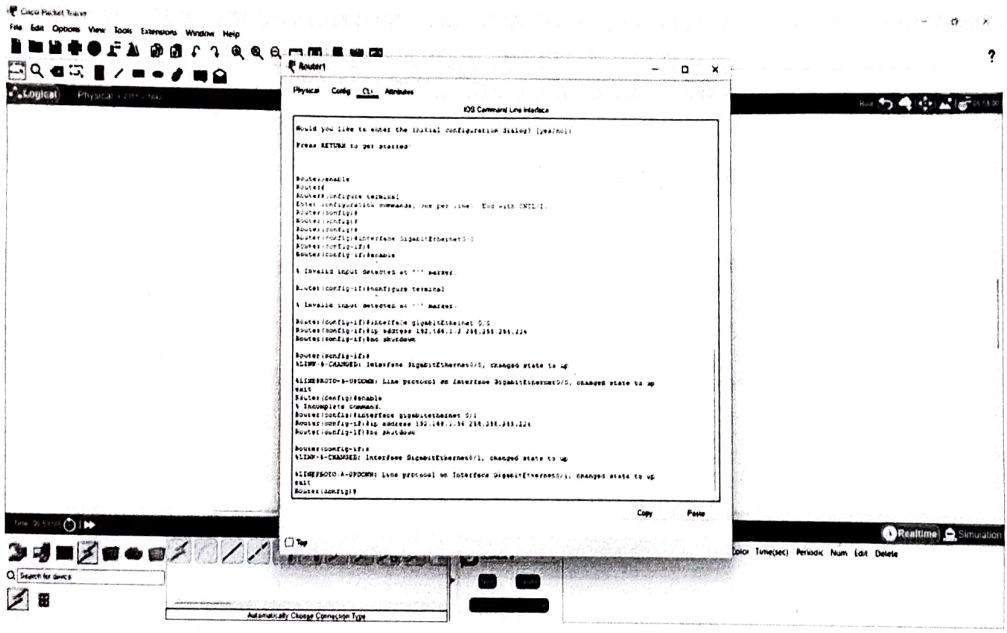
Result:

The peer to peer hub if switch network were successfully assigned and observed.


Router1
PC0
PC1




Cisco Packet Tracer
Logical
Physical
PC-PT
PC0
PC-PT
PC1
Realtime
Simulation
Automatically Choose Connection Type


Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical
Physical
Router1
Physical Config CLI Attributes
IOS Command Line Interface
Would you like to enter the initial configuration dialog? [yes/no]:
Press RETURN to get started.
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#
Router(config)#
Router(config)#interface GigabitEthernet0/0
Router(config-if)#
Router(config-if)#noshut
% Invalid input detected at '^' marker.
Router(config-if)#configure terminal
% Invalid input detected at '^' marker.
Router(config-if)#interface gigabitEthernet 0/0
Router(config-if)#ip address 192.168.1.2 255.255.255.224
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
exit
Router(config)#enable
% Incomplete command.
Router(config)#interface gigabitEthernet 0/1
Router(config-if)#ip address 192.168.1.34 255.255.255.224
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
exit
Router(config)#
Copy
Paste
Top
Realtime
Simulation
Automatically Choose Connection Type

